

DESIGN AND ANALYSIS OF ALGORITHMS

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'B' SECTION CSE RGCT

TRAVELLING SALEMAN PROBLEM

- Let $G(V, E)$ be a directed graph with edge cost C_{ij} is defined such that $C_{ij} > 0$ for all i and j and $C_{ij} = \infty$, if $\langle i, j \rangle \notin E$.
- Let $V = n$ and assume $n > 1$.
- The Travelling salesman problem is to find a tour of minimum cost.
- A tour of G is a directed cycle that include every vertex in V .
- The cost of the tour is the sum of cost of the edges on the tour.
- The tour is the shortest path that starts and ends at the same vertex (ie) 1.

APPLICATION

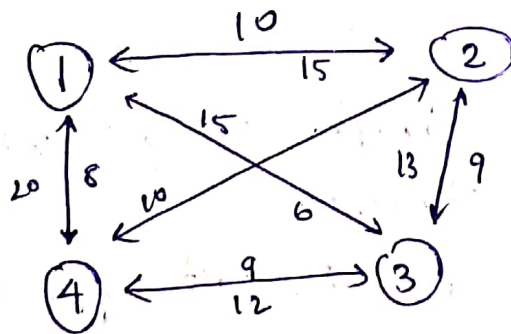
- An $n+1$ vertex Graph can be used to represent the situation.
- One vertex represents the post office from which the postal van starts and return.
- Edge $\langle i, j \rangle$ is assigned a cost equal to the distance from site 'i' to site 'j'.
- Every tour consists of an edge $\langle k, 1 \rangle$ for some $k \in V - \{1\}$ and a path from vertex k to vertex 1.
- The path from vertex k to vertex 1 goes through each vertex in $V - \{1, k\}$ exactly

one.

- The function which is used to find the path is $g(i, V - \{i\}) = \min \{C_{ij} + g(j, S - \{j\})\}$
- The function $g(i, V - \{i\})$ is the length of an optimal tour.

STEPS TO FIND THE PATH

- 1) Find $g(i, \emptyset) = C_{ii}$, $1 \leq i \leq n$, hence we can use eqn (2) to obtain $g(i, S)$ for all S to size 1.
- 2) That we have to start with $S = 1$, (ie). there will be only one vertex in set 'S'.
- 3) Then $S = 2$ and we have to proceed until $|S| \leq n-1$.
- 4) For example consider the graph.



Cost matrix

0	10	15	20
5	0	9	10
6	13	0	12
8	8	9	0

$g(i, s) \rightarrow$ set of nodes / vertex have to
 ↓ visited: s

starting position $g(i, s) = \min \{c_{ij} + g(j, s - \{i\})\}$

Step 1:

$$\begin{aligned} g(1, \{2, 3, 4\}) &= \min \{c_{12} + g(2, \{3, 4\}), \\ &\quad c_{13} + g(3, \{2, 4\}), c_{14} + g(4, \{2, 3\})\} \\ &= \min \{10 + 25, 15 + 25, 20 + 23\} \\ &= \min \{35, 35, 43\} \\ &= 35 \end{aligned}$$

Step 2:

$$\begin{aligned} g(2, \{3, 4\}) &= \min \{c_{23} + g(3, \{4\}), c_{24} + g(4, \{3\})\} \\ &= \min \{9 + 20, 10 + 15\} \\ &= \min \{29, 25\} \\ &= 25 \end{aligned}$$

$$\begin{aligned} g(3, \{2, 4\}) &= \min \{c_{32} + g(2, \{4\}), c_{34} + g(4, \{2\})\} \\ &= \min \{13 + 18, 12 + 13\} \\ &= \min \{31, 25\} \\ &= 25 \end{aligned}$$

$$\begin{aligned} g(4, \{2, 3\}) &= \min \{c_{42} + g(2, \{3\}), c_{43} + g(3, \{2\})\} \\ &= \min \{8 + 15, 9 + 18\} \\ &= \min \{23, 27\} \\ &= 23 \end{aligned}$$

Step 3:

$$g(3, \{4\}) = \min \{c_{34} + g(4, \emptyset)\}$$

$$\Rightarrow 12 + 8 = 20$$

$$g(4, \{3\}) = \min \{c_{43} + g(3, \emptyset)\}$$

$$\Rightarrow 9 + 6 = 15$$

$$g(2, \{4\}) = \min \{c_{24} + g(4, \emptyset)\}$$

$$g(3, \{2\}) = \min \{c_{32} + g(2, \emptyset)\}$$

$$13 + 5 = 18$$

$$|S| = 0$$

$i = 1$ to n

$$g(1, \emptyset) = c_{11} \Rightarrow 0$$

$$g(2, \emptyset) = c_{21} \Rightarrow 5$$

$$g(3, \emptyset) = c_{31} \Rightarrow 6$$

$$g(4, \emptyset) = c_{41} \Rightarrow 8$$

$$|S| = 1$$

$i = 2$ to 4

$$g(2, \{3\}) = c_{23} + g(3, \emptyset) = 9 + 6 = 15$$

$$g(2, \{4\}) = c_{24} + g(4, \emptyset) = 10 + 8 = 18$$

$$g(3, \{2\}) = c_{32} + g(2, \emptyset) = 13 + 5 = 18$$

$$g(4, \{2\}) = c_{42} + g(2, \emptyset) = 8 + 5 = 13$$

$$g(4, \{3\}) = c_{43} + g(3, \emptyset) = 9 + 6 = 15$$

$$S = \{2\}$$

$$i \neq 1, i \in S \text{ and } j \in S$$

$$g(2, \{3, 4\}) = \min \{c_{23} + g(3, \{4\}), c_{24} + g(4, \{3\})\}$$

$$= \min \{9 + 20, 10 + 15\}$$

$$= \min \{29, 25\}$$

$$= 25$$

$$g(3, \{2, 4\}) = \min \{c_{23} + g(3, \{4\}), c_{24} + g(4, \{3\})\}$$

$$= \min \{9 + 20, 10 + 15\}$$

$$= \min \{29, 25\}$$

$$= 25$$

$$g(3, \{2, 4\}) = \min \{c_{32} + g(2, \{4\}), c_{34} + g(4, \{2\})\}$$

$$= \min \{13 + 19, 12 + 13\}$$

$$= \min \{31, 25\}$$

$$= 25$$

$$g(4, \{2, 3\}) = \min \{c_{42} + g(2, \{3\}), c_{43} + g(3, \{2\})\}$$

$$= \min \{8 + 15, 9 + 18\}$$

$$= \min \{23, 27\}$$

$$= 23$$

$$|S| = 3$$

$$g(1, \{2, 3, 4\}) = \min \{c_{12} + g(2, \{3, 4\}), c_{13} + g(3, \{2, 4\}), c_{14} + g(4, \{2, 3\})\}$$

$$= \min \{10+25, 18+25, 20+23\}$$

$$= \min \{25, 35, 43\}$$

$$= 35$$

Optimal cost is 35

The shortest path is -

$$g(1, \{2, 3, 4\}) = C_{12} + g(2, \{4, 3\}) = 1 \rightarrow 2$$

$$g(2, \{3, 4\}) = C_{24} + g(4, \{3\}) = 1 \rightarrow 2 \rightarrow 4$$

$$g(4, \{3\}) = C_{43} + g(3, \{\emptyset\}) = 1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 1$$

So the optimal tour is $1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 1$

BICONNECTED COMPONENTS

Articulation point

A vertex 'v' in an connected graph 'G' is an articulation point, if the deletion of vertex v together with all vertex incident to v disconnects the graph into 2 or more non empty components.

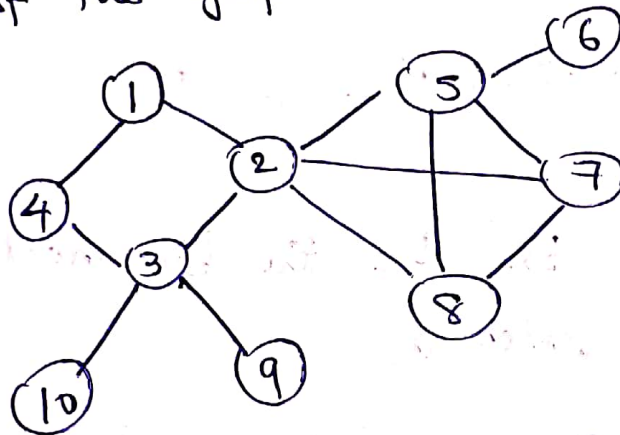
Bi connected Graph

A Graph 'G' is biconnected, if it contains no articulation points.

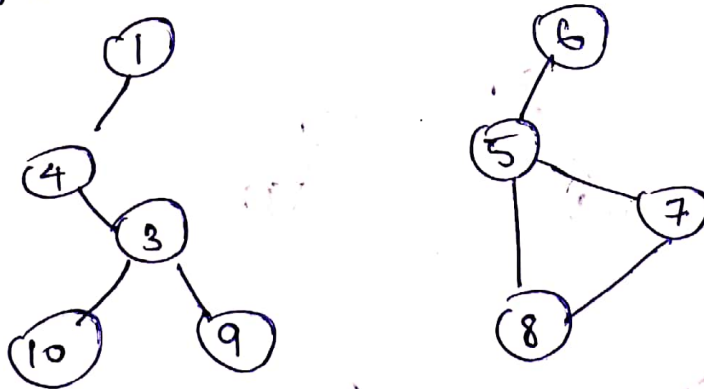
Example

'G' is a graph representing a communication network, vertices represent communication station, edge represent communication lines.

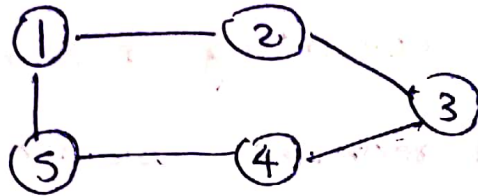
- If communication station has articulation point.
- If the graph has no articulation point.



In the above graph G , if we delete the vertex 2, then it creates 2 non empty components.



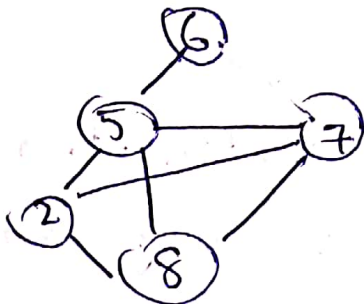
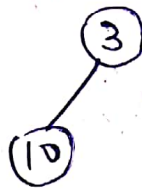
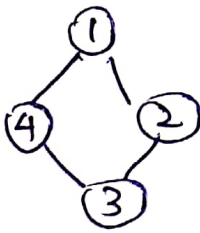
If we delete any other vertex other than 2, 3 and 5, then we get only one component exactly. So the graph is not biconnected.



The above graph is biconnected. Because it has no articulation point.

Algorithm to construct A biconnected Graph

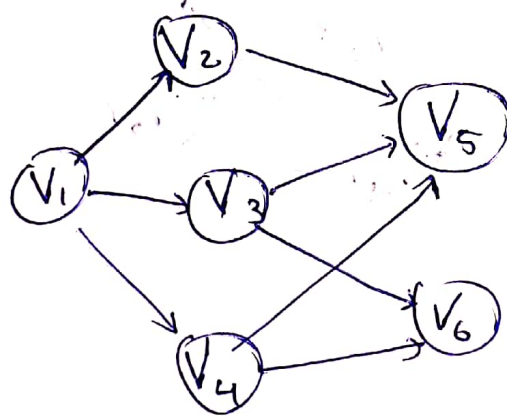
1. For each articulation point, 'a' do.
2. {
3. Let B_1, B_2, B_k be the biconnected component containing vertex a;
4. Let $v_i, v_{i+1} \neq a$ be a vertex in $B_i, 1 \leq i \leq k$.
5. add to 'G' the edge $(v_i, v_{i+1}), 1 \leq i \leq k$;
6. }



TOPOLOGICAL SORTING

A directed graph G in which the vertices represent tasks / events and the edges represent activities to move from one event to another. Then that graph is known as activity on vertex network / AOV - network / PERT graph.

PERT graph is made to analyze interrelated activities of complete projects. The purpose of a Topological sort is to order the events in a linear manner with the restriction that an event cannot precede other events that must first take place.



$V_1, V_2, V_5, V_3, V_6, V_4$

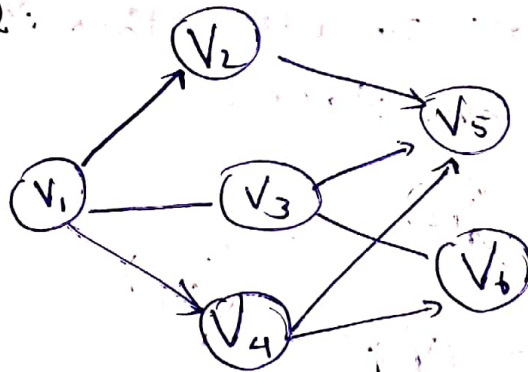
$V_1, V_4, V_3, V_6, V_2, V_5$

Topological order

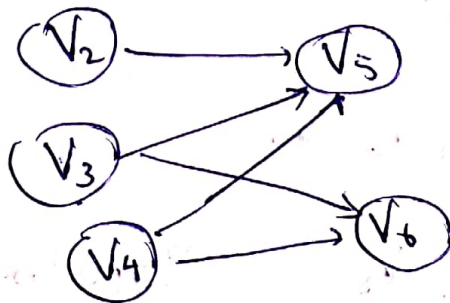
ALGORITHM

- Let n be the no. of vertices
1. For $i \rightarrow 1$ to n do
 2. If every vertex has no predecessor then
 3. pick a vertex
 4. O/p v
 5. Delete v and all edge leading and of v from network
 6. End.

a) Initial:



b) V_1



Topological order is V_1, V_4, V_3, V_2, V_5 .

In this function are

- This can be efficiently done if for each vertex a count of the no of its immediate predecessor is kept.
- This can be implemented if the network is represented by its adjacency lists.
- Decide whether a vertex has any predecessor
- Decide a vertex together with all its incident edges

Algorithm

1. $top \leftarrow 0$
2. for $i \leftarrow 1$ to n do
3. if $count(i) = 0$ then
4. end
5. for $i \leftarrow 1$ to n do
6. if $top = 0$ then
7. $j \leftarrow top$; $top \leftarrow count(top)$;
8. $ptr \leftarrow LINK(j)$
9. While $ptr \neq \emptyset$ do

10. // decrease the count of successor vertices of j
11. $K \leftarrow \text{VERTEX}(\text{ptr})$
12. $\text{COUNT}(K) \leftarrow \text{COUNT}(K) - 1$
13. If $\text{COUNT}(K) = 0$
14. Then, $\text{COUNT}(K) \leftarrow \text{top}; \text{top} \leftarrow K]$
15. $\text{ptr} \leftarrow \text{LINK}(\text{ptr})$
16. end
17. End
18. End TOPOLOGICAL $\rightarrow$ Order.